

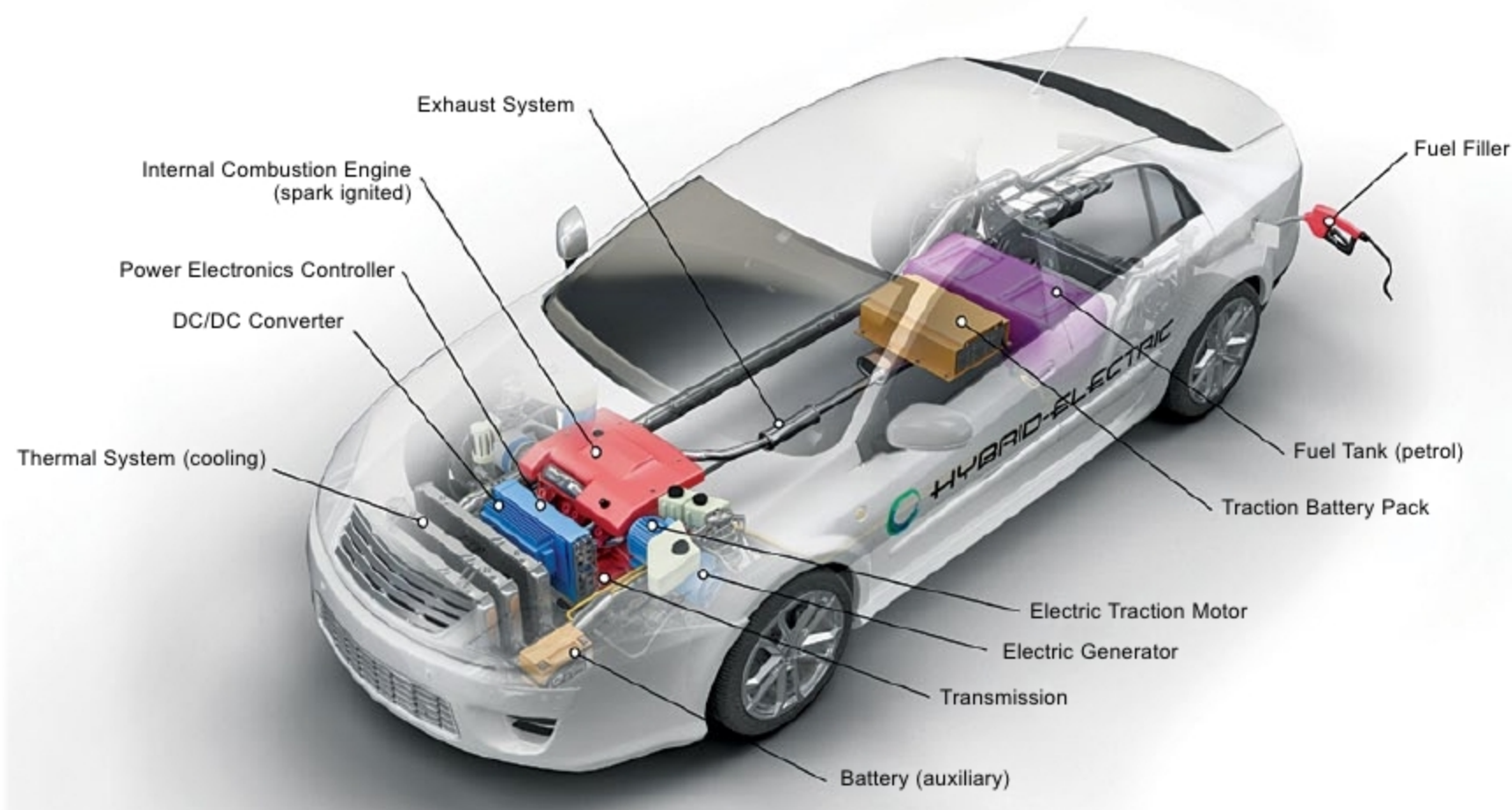
Evaluation Of ELECTRIC VEHICLES For Performance, Safety And EMC

Electric vehicles (EVs) are often seen as the green transport option. As the automotive industry advances in power conversion and battery technologies, it also improves EV testing performance and cost by developing flexible automated power conversion test platforms and regenerative battery test systems

EFY BUREAU

The electric vehicle (EV) industry is constantly improving its technologies to reduce carbon footprint and meet customers' expectations for a good driving experience and a reasonable recharge time. These factors are driving EV manufacturers to develop efficient power converters, inverters, alternators, faster charging techniques/batteries and improved battery performances. EV supply equipment (EVSE) is evolving to encompass quick-charging batteries, public charging stations and sophisticated onboard components that enhance battery performance and ensure future EV compatibility with EVSE infrastructures.

If EVSE components are not evaluated properly, these can cause battery failures and danger to vehicle operators in hazardous conditions. To deliver safe and high-performing EVs, several regulation standards are established. Safety test applications include motors, power systems, batteries, charging systems, wiring and con-



Electric vehicle supply equipment (Credit: <https://afdc.energy.gov>)